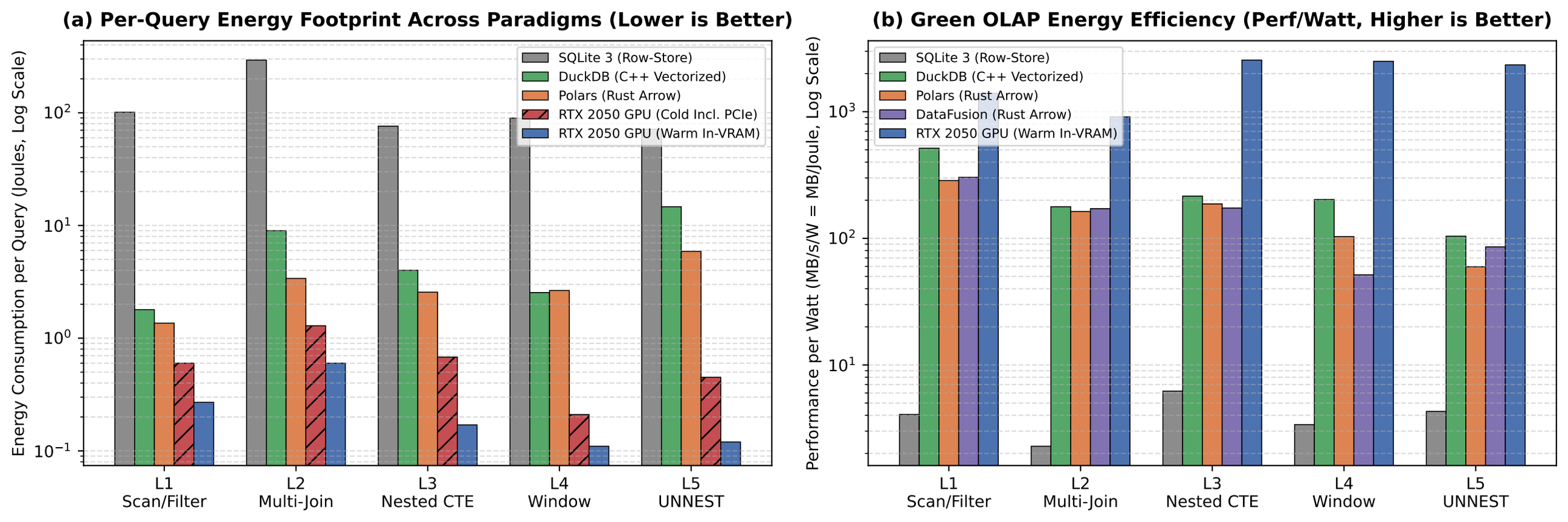
(a) Per-Query Energy Footprint Across Paradigms (Lower is Better)
Energy Consumption per Query (Joules, Log Scale)
SQLite 3 (Row-Store)
DuckDB (C++ Vectorized)
Polars (Rust Arrow)
RTX 2050 GPU (Cold Incl. PCIe)
RTX 2050 GPU (Warm In-VRAM)
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST
(b) Green OLAP Energy Efficiency (Perf/Watt, Higher is Better)
Performance per Watt (MB/s/W = MB/Joule, Log Scale)
SQLite 3 (Row-Store)
DuckDB (C++ Vectorized)
Polars (Rust Arrow)
DataFusion (Rust Arrow)
RTX 2050 GPU (Warm In-VRAM)
L1 Scan/Filter
L2 Multi-Join
L3 Nested CTE
L4 Window
L5 UNNEST